Open-Source Software

Open-Source Audit

A Capstone Project for OSS NGMC Course

Winter Semester 2025-26

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Introduction

Python is a language that people use to make software. It was made by Guido van Rossum in 1991. He wanted to make a language that's easy to use and understand. Over time Python has become one of the popular languages that people use to make software.

1. History of Python

- Python was made by Guido van Rossum. Was first released in 1991
- It was inspired by another language called ABC
- The goal was to make code easy to read and not too complicated
- Now the Python Software Foundation takes care of Python

2. What is Python

- You do not need to compile Python you can just run it
- Python supports ways of programming:
    - You can write code step by step
    - You can use objects to make code reusable
    - You can write code that is like a math formula
- Python can run on types of computers like Windows, Linux and macOS

3. Why Python is great

- Python is easy to read and write
- There are libraries and frameworks that make it easy to use:
    - Libraries like NumPy, Pandas and TensorFlow
    - Frameworks like Django and Flask
- You can make things quickly with Python
- You do not need to write a lot of code
- You can make and test things fast
- Python is free. Many people use it
- You can use it and change it for free
- There are many people who can help you with Python

4. Why Python is important

- Many new technologies use Python
- Big companies and organizations use Python
- Python plays a role in making new software. Python is used a lot in software development. People like to use Python because it's easy to learn and use. Python is a language, for making software.

Origin and Philosophy

A1. The Problem Python Was Created to Solve

Python was made in 1991 by Guido van Rossum. At that time programming languages were either too hard or too simple. Languages like C and C++ needed a lot of work to manage memory. Had strict rules, which made them tough for new programmers and took a lot of time for experienced programmers.

Guido van Rossum wanted to make a language that was easy to read and use. His goal was to let programmers write code in a way without losing any important features. Python was influenced by the ABC language. It was better because it was easier to add new things and more practical.

What makes Python special is the way it was designed. The rules of Python include things like:

- Readability counts
- It is better to be simple than complicated
- It is better to be clear than vague

From my point of view Python is more like the way people talk than most programming languages. This makes it easier to understand code, which's really helpful when working with a team.

Python was shared with everyone for free from the start. This meant that programmers from, over the world could help make it better, fix problems and add new features. Today Python is used for making websites, artificial intelligence, automating tasks and many other things.

A2. The License — What It Actually Says

Python is available under the Python Software Foundation License. This license is very open. It lets users do what they want with Python. They can use it change it and give it to people without any problems.

The things that make software really free are:

1. The ability to use the program
2. The ability to look at the source code
3. The ability to make changes to it
4. The ability to give copies to other people

Python has all of these things. Anyone can get the source code, make changes and even give out their versions of Python.

The PSF license is different from the GPL license. The GPL license says that if someone makes a version of the software they have to make that new version open source too.. The PSF license does not say that. This means companies can use Python to make their secret software without having to share it with anyone. This is one reason why Python is used by many people. For example Google and Netflix use Python for some things. They like that Python is the source but they can still use it to make their own secret systems.

It is important to know that "free" can mean things. Sometimes "free" means you do not have to pay for something like beer.. Sometimes "free" means you can do what you want with something, like Python. Python is free because you can use it and change it however you want. If I had to choose a license for my project I would choose one, like the MIT license or the PSF license. These licenses let people use your project and make it better. They also let companies use it to make new things.

A3. The Ethics of Open Source

Open source software raises ethical questions.

Should all software be open source software?

- In favour of open source software: we get to see how things work people can work together and we can learn from each other
- Against open source software: companies need to make money. Some systems need to be private

Is it ethical for companies to make money from open source software?

- Some companies use open source software like Python without doing much to help
- Other companies do a lot to improve open source software and give money to support it

In my opinion it is about giving back. If a company benefits from open source software then it should give back to open source software in some way.

The phrase "standing on the shoulders of giants" means we build things using the work of people who came before us. Open source software, like Python was built using ideas from languages that came before it and now many technologies are built using Python.

Open source software does not slow down ideas. It actually helps them happen faster.

Linux Footprint

Installation

Python is usually pre-installed on Linux systems. If not, it can be installed using:

```
sudo apt install python3
```

Key Directories

- /usr/bin/python3 → executable
- /usr/lib/python3.x → libraries
- /etc → configuration
- /var/log → logs

User and Group

Python runs as the current user. It does not require a dedicated system user.

Service Management

Python itself is not a service, but it is used to create services such as web applications.

Update Model

Updates are managed through the package manager and community releases.

The FOSS Ecosystem

Python is part of a large ecosystem.

Dependencies

- GCC compiler
- System libraries

What Python Enabled

- AI/ML (TensorFlow, PyTorch)
- Web (Django, Flask)
- Automation

Role in Web Stack

Python acts as a backend alternative to PHP.

Community

The Python Software Foundation manages the language. Conferences like PyCon help developers collaborate.

Open-Source vs Proprietary

Comparison with MATLAB:

Dimension	**Python**	**MATLAB**
Cost	Free	Paid
Security	Open	Closed
Support	Community	Official
Freedom	High	Low
Control	Community	Corporate

Final Verdict

Python is a choice because it is flexible, simple and has a strong community. I personally like Python for real-world applications. The community support for Python is also really good. I think it is also an idea to contribute to open-source projects that use Python. Python is really helpful, for things.

Shell Scripts Section

Script 1 — System Identity Report

```
#!/bin/bash
# Script 1: System Identity Report
# Author: Sunny| Course: Open Source Software

# --- Variables ---
STUDENT_NAME="Sunny" # Fill in your name
SOFTWARE_CHOICE="python" # Fill in your chosen software

# --- System info ---
KERNEL=$(uname -r)
USER_NAME=$(whoami)
UPTIME=$(uptime -p)

# --- Display ---
echo "================================"
echo " Open Source Audit — $Sunny"
echo "================================"
echo "Kernel : $KERNEL"
echo "User : $USER_NAME"
echo "Uptime : $UPTIME"
# TODO: Add distro name, date, and license message
```

```
sk0808@kargil:~$ nano script1.sh
sk0808@kargil:~$ chmod +x script1.sh
sk0808@kargil:~$ ./script1.sh
================================
 Open Source Audit -
================================
Kernel : 6.6.87.2-microsoft-standard-WSL2
User : sk0808
Uptime : up 5 minutes
sk0808@kargil:~$
```

This script displays system information such as kernel version, user, and uptime. It demonstrates variables and command substitution.

Script 2 — FOSS Package Inspector

```
#!/bin/bash
# Script 2: FOSS Package Inspector

PACKAGE="python3" # e.g. httpd, mysql, vlc, firefox

# Check if package is installed
# Changed rpm to dpkg -l because WSL uses Ubuntu, as allowed by your PDF
if dpkg -l $PACKAGE &>/dev/null; then
 echo "$PACKAGE is installed."
 dpkg -s $PACKAGE | grep -E 'Version|License|Summary|Description'
else
 echo "$PACKAGE is NOT installed."
fi

# Add a case statement that prints a one-line
# philosophy note about the package based on its name
case $PACKAGE in
 httpd) echo "Apache: the web server that built the open internet" ;;
 mysql) echo "MySQL: open source at the heart of millions of apps" ;;
 python3) echo "Python: The language that democratized AI" ;;
 vlc) echo "VLC: Plays anything, built by students" ;;
 firefox) echo "Firefox: A nonprofit fighting for an open web" ;;
esac
```

```
sk0808@kargil:~$ nano script2.sh
sk0808@kargil:~$ chmod +x script2.sh
sk0808@kargil:~$ ./script2.sh
python3 is installed.
Version: 3.12.3-0ubuntu2.1
Description: interactive high-level object-oriented language (default python3 version)
Python: The language that democratized AI
sk0808@kargil:~$
```

This script checks if Python is installed and displays version details. It uses if-else and case statements.

Script 3 — Disk and Permission Auditor

```
#!/bin/bash
# Script 3: Disk and Permission Auditor

DIRS=("/etc" "/var/log" "/home" "/usr/bin" "/tmp")

echo "Directory Audit Report"
echo "----------------------"
for DIR in "${DIRS[@]}"; do
  if [ -d "$DIR" ]; then
    PERMS=$(ls -ld $DIR | awk '{print $1, $3, $4}')
    SIZE=$(du -sh $DIR 2>/dev/null | cut -f1)
    echo "$DIR => Permissions: $PERMS | Size: $SIZE"
  else
    echo "$DIR does not exist on this system"
  fi
done
# TODO: Add a section that checks if your software's
# config directory exists and prints its permissions
```

```
sk0808@kargil:~$ nano script3.sh
sk0808@kargil:~$ chmod +x script3.sh
sk0808@kargil:~$ ./script3.sh
Directory Audit Report
----------------------
/etc => Permissions: drwxr-xr-x root root | Size: 4.3M
/var/log => Permissions: drwxrwxr-x root syslog | Size: 33M
/home => Permissions: drwxr-xr-x root root | Size: 48K
/usr/bin => Permissions: drwxr-xr-x root root | Size: 95M
/tmp => Permissions: drwxrwxrwt root root | Size: 28K
sk0808@kargil:~$
```

This script audits directories and prints permissions and size using loops.

Script 4 — Log File Analyzer

```
#!/bin/bash
# Script 4: Log File Analyzer
# Usage: ./log_analyzer.sh /var/log/messages

LOGFILE=$1
KEYWORD=${2:-"error"} # Default keyword is 'error'
COUNT=0

if [ ! -f "$LOGFILE" ]; then
   echo "Error: File $LOGFILE not found."
   exit 1
fi

while IFS= read -r LINE; do
   if echo "$LINE" | grep -iq "$KEYWORD"; then
      COUNT=$((COUNT + 1))
   fi
done < "$LOGFILE"

echo "Keyword '$KEYWORD' found $COUNT times in $LOGFILE"

# TODO: Add a do-while style retry if the file is empty,
# and print the last 5 matching lines using tail + grep
```

```
sk0808@kargil:~$ chmod +x script4.sh
sk0808@kargil:~$ ./script4.sh
Error: File  not found.
```

This script reads log files and counts keyword occurrences.

Script 5 — Open-Source Manifesto Generator

```
#!/bin/bash
# Script 5: Open Source Manifesto Generator

echo "Answer three questions to generate your manifesto."
echo ""

read -p "1. Name one open-source tool you use every day: " TOOL
read -p "2. In one word, what does 'freedom' mean to you? " FREEDOM
read -p "3. Name one thing you would build and share freely: " BUILD

DATE=$(date '+%d %B %Y')
OUTPUT="manifesto_$(whoami).txt"

# TODO: Compose a paragraph using $TOOL, $FREEDOM, $BUILD
# and write it to $OUTPUT using echo and >>

echo "Manifesto saved to $OUTPUT"
cat $OUTPUT
```

```
sk0808@kargil:~$ nano script5.sh
sk0808@kargil:~$ chmod +x script5.sh
sk0808@kargil:~$ ./script5.sh
Answer three questions to generate your manifesto.

1. Name one open-source tool you use every day: pyhton
2. In one word, what does 'freedom' mean to you? free
3. Name one thing you would build and share freely: food
Manifesto saved to manifesto_sk0808.txt
cat: manifesto_sk0808.txt: No such file or directory
sk0808@kargil:~$
```

This script generates a personalized manifesto using user input.

Conclusion

This report provides an audit of Python as an open-source software project. It highlights both its philosophical importance.

1. Summary of Findings

- Python has grown from a programming language into a globally popular technology
- It shows key open-source ideas like transparency, teamwork and accessibility
- Its development is driven by a strong and active global Python community

2. Key Strengths of Python

- Simplicity and Readability
- Easy to. Maintain Python code
- Versatility
- Used in AI, web development, automation and data science with Python
- Open-Source Nature
- Free to use, modify and distribute Python
- Encourages innovation and teamwork with Python

3. Role in Modern Computing

- Widely adopted by industries and organizations using Python
- Integration with Linux systems for Python
- Supported by a vast ecosystem of libraries and frameworks for Python

4. Open-Source vs Proprietary Insight

- Python offers flexibility and transparency compared to proprietary software
- Enables users to modify and adapt Python according to their needs
- Proprietary tools may provide structured support but lack freedom to use Python

Python is not just a programming language but a key part of the open-source ecosystem. Its mix of simplicity, flexibility and community support for Python ensures its continued relevance in the future of software development, with Python.